



Band aid for organs

The easily detachable double sided surgical tape developed by MIT researchers over years. <https://digit.in/aug20-11>



Surgical adhesives

An overview of the efficacy and use cases of the many conventional surgical adhesives in use today. <https://digit.in/aug20-12>

the body. To do this, they used gelatin, or chitosan, which is a hard substance found in the shells of insects. By infusing the adhesive with these substances, the shape of the tape is retained within the body even after the surgical procedure is complete. By varying the concentration of these substances within the adhesive, the researchers can control how long the tape can last. While gelatin dissolves within a few days or weeks, the chitosan can last many months, even a year if necessary. Apart from the surgical procedures, the tape can also be used to attach medical devices and sensors on the surface of organs.

There was however, a problem with the tape. The NHS esters formed such strong bonds that the underlying tissue could be damaged if it was necessary to subsequently remove the tape. To tackle this problem, the researchers first modified the chemical mixup in the original tape, and included disulfide, a substance that forms strong bonds but can be easily removed with a reducing substance. That substance was a biocompatible antioxidant known as glutathione,

ing heart, moving lungs or pulsing blood vessels, an international team of researchers from many universities have created hydrogel-based adhesives that can be simply squirted on to the required area. Normally, these organs are always moving, and the motion can cause further stress on the organs depending on the method used to close the wound. The constant motion also means that more conventional adhesives are likely to simply be ineffective or wear out quicker. The required solution has to be incredibly elastic, and strong at the same time.

The range of hydrogel adhesives that promise to solve the problem, called MeTro are all developed for particular applications within the body, and use a range of substances. A seaweed extract used to thicken foodstuff, known as alginate as well as the main material for soft contact lenses called

polyacrylamide are both used to make hydrogel adhesives. Another hydrogel adhesive uses elastin for the sealant which is the same material naturally found in elastic tissues such as veins and lungs. Chitosan is also used to give tensile strength to these adhesives.

These hydrogels stay in the place of application without mov-

ing within the body. Additionally, the formulations can be "cured" with UV light which makes them solidify in place stronger, making the procedure more effective. The potential applications for these adhesives are wide ranging, from the emergency wards to treat automobile accidents, to the injuries of soldiers in war zones. The

ease of application and the fast action helps in both these situations.

In the stress tests for MeTro adhesives, they were shown to outperform existing adhesives and stretch up to 20 times its length before breaking. They can withstand up to three times the stress of conventional medical adhesives before breaking.



Credit: University of Sydney

The Squirtable MeTro glue

The delicate organs are often soaked in liquid or blood, which makes the action of conventional adhesives difficult. Existing adhesives also work in tandem with staples or sutures, and do not work at all without them. The MeTro adhesives however are strong enough to work in these demanding environments, independently of support from other tissue mending procedures. When treated with the UV light, MeTro sets within a minute, which is incredibly fast for such strong adhesives. The process is similar to the silicon sealants used to repair plumbing in bathrooms and kitchens, but for the human body. Before the curing process, MeTro behaves like a viscous liquid filling in the gaps and moulding itself to the surface of the tissue. For healing cuts, the procedure can be as simple as squirting the MeTro gel onto the open wound with a syringe. The gel stays in place for as long as the delicate organs that it is applied on require to heal. When it eventually degrades, it does so without any toxicity in the human body. MeTro adhesives have already been tested on rats and pigs, and are moving to human trials. 

Credit: Felice Frankel, Christine Daniloff, MIT



Double sided adhesive tape developed by MIT for holding tissue together.

which is found in human bodies naturally. With this combination, the tape could be removed when required. The tape has been tested on pig hearts, intestines and lungs.

METRO

To tackle the problem of keeping together fragile tissues like the breath-